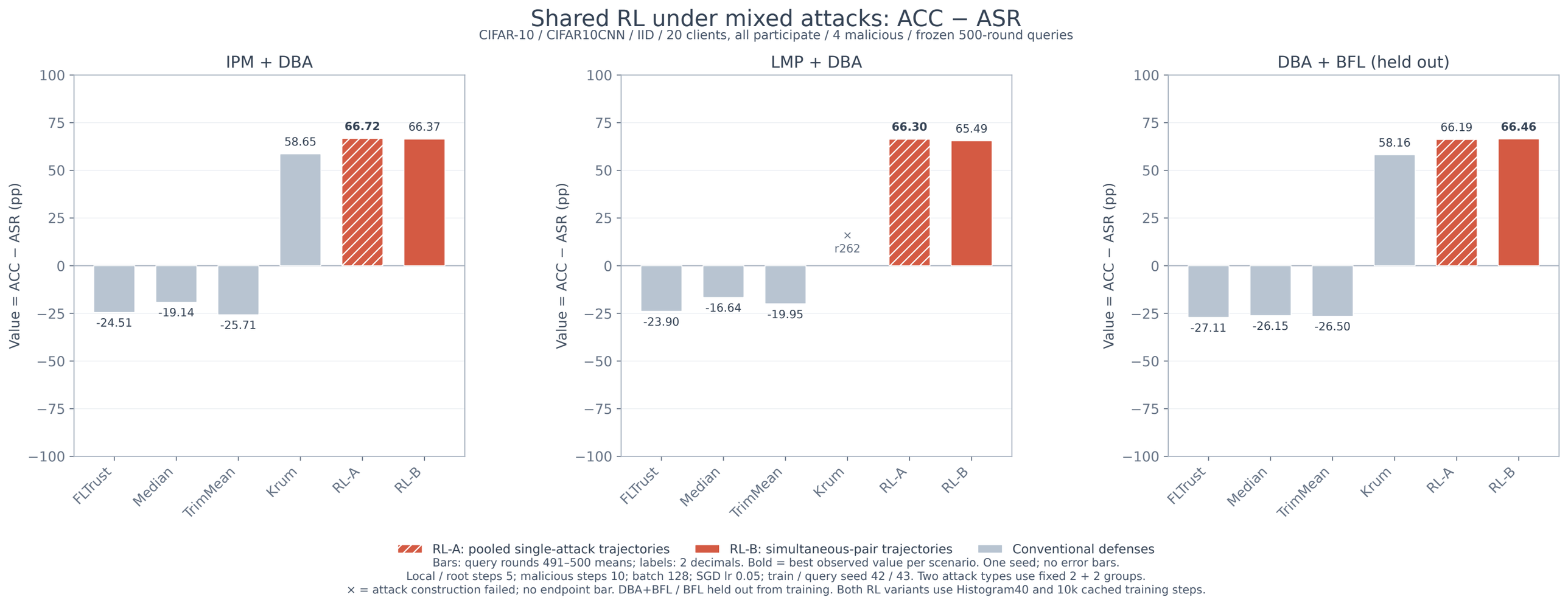
Shared RL under mixed attacks: ACC − ASR
CIFAR-10 / CIFAR10CNN / IID / 20 clients, all participate / 4 malicious / frozen 500-round queries
IPM + DBA
LMP + DBA
DBA + BFL (held out)
Value = ACC − ASR (pp)
100
75
50
25
0
−25
−50
−75
−100
-24.51
-19.14
-25.71
58.65
66.72
66.37
-23.90
-16.64
-19.95
×
r262
66.30
65.49
-27.11
-26.15
-26.50
58.16
66.19
66.46
FLTrust
Median
TrimMean
Krum
RL-A
RL-B
RL-A: pooled single-attack trajectories
RL-B: simultaneous-pair trajectories
Conventional defenses
Bars: query rounds 491–500 means; labels: 2 decimals. Bold = best observed value per scenario. One seed; no error bars.
Local / root steps 5; malicious steps 10; batch 128; SGD lr 0.05; train / query seed 42 / 43. Two attack types use fixed 2 + 2 groups.
× = attack construction failed; no endpoint bar. DBA+BFL / BFL held out from training. Both RL variants use Histogram40 and 10k cached training steps.